

ActInf Textbook Group ~ Cohort 2 ~ Meeting 11

(Chapter 5, part 2)

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hello everyone thanks for joining or listening it's November 11th 11 11 22. and we're in cohort two of the active textbook group having a discussion of chapter five exploring some topics there are some questions that have been written in comments as well and then taking it wherever we want as we review the first half of the book that'll be today next week we'll be taking an even further step back and more of a focus on like feedback completing the feedback form how can we improve and structure future textbook group cohorts project ideas and so on but for today we're in chapter five so um does anyone want to give any thoughts or comments or just initial questions to raise about chapter five all right then let's go into the written questions okay how does the simple hierarchy with one-to-one mapping between levels as shown in the left hand side of figures 4.6. and 5.2 four six five two relate to actual brain hierarchies such as the classic felleman and Van Essen diagram from retinal ganglion cells up to hippocampus with many to many mapping between Regional units ah I guess this is the famous well what do people think how do these seemingly skyscraper like hierarchical models this one is laid out side to side but we can see ascending and descending so this could have been graphically laid out like this and then another one and then a third one on top of course top down the bottom-up are just spatial metaphors how do we reconcile computational hierarchical models with the empirical structural connectivity of the brain which has a wiring structure looking like this Ali between these two kinds of icons or kinds of maps because on one hand we have a kind of relational map and on the other hand there is a kind of architectural slash functional mapping of the brain so for example if we look at a circuit map of the of an electric circuit there's we have a relational map there right but when we want to implement that circuit I mean in terms of wiring and so on we need to have another kind of map which is architectural one of course in some cases there's a possibility of translating one map perfectly onto the other but there isn't necessarily an isomorphism between those two kinds of map and I think in this case we don't necessarily see that isomorphism here as well

excellent great answer totally agreed these kinds of diagrams are often described as hierarchical or heterotical or people use difference words to describe it but it's describing the anatomical connectivity of different brain regions whereas the diagrams that are shown in these figures that were mentioned in the question are not anatomical Regions they're statistical variables connected by edges this is the difference between structural and effective and functional connectivity structural connectivity is what pieces are actually put together and through direct contact intermediating whereas functional connectivity has to do with the statistical relationships in the system and those are not necessarily for a given set of measurements or spatial or temporal scale or experimental stimuli those are not exactly the same thing there are cases where there are but that doesn't mean that in general they're the same thing so how do they relate to quote actual brain hierarchies they relate to each other in the way that Ali just described similarly between um uh architectural diagram and a functional diagram does anyone have any other thoughts on this yeah Giuseppe yeah I think that is a is a very good explanation on the other hand I think that at that stage of the of the description the idea was to see how active inference could also provide a process Theory so basically to to see how these information mechanism may be implemented in a neural circuits so yes the two are not completely the two descriptions are not isomorphic but I think there should be some degree of of uh agreement and in this case I think that what may be masking the the the similarities or the the potential overlap is that you know there there's a a big difference in the degree of complexity of the so one is a very simple the the active inference diagram is a very simple sketch of a very simple micro circuit while this one is a big architecture with a lot of complexity and it's difficult to to go in and say you know to map exactly those kind of micro circuits within this large circuit scheme it's probably the there's going to be various micro circuits in there with the various degrees of a hierarchical structure within that that large scheme I guess yeah um all great points the figures that were pointed to reflect like just simple illustrative examples whereas the whole brain or many of the regions of the brain if one were to even just describe this with an active inference model it wouldn't look as simple and narrow as these previous figures so it's a different type of diagram it's a different scale yet through Visual and prose-based argumentation as well as the the hopes and dreams of a field is that there is a relationship between the messages that are passed between different variables statistically which as we discussed last time it can you could arrange these in a hexagon it wouldn't change the topology and thus the function of the base graph however it is being laid out in a specific way to demonstrate resonance with given cellular populations in a cortical

column so it's kind of situational to what extent the structural neuroanatomy aligns with the Bayes graph as specified but we can definitely say that they're not the same thing that they don't have to be the same thing but there's also situations where by graphical design using that in multiple ways meaning you could constrain your construction of the base graph such that it has some structural patterns with the anatomy and then also just in terms of like visual graphical design like you can just lay it out visually because this is not these are it's not exactly like there's five and then three like there's other things that are here that aren't shown here so like this is a a course mapping any other thoughts on this question asked uh just really briefly I feel like the 30 million foot view of the like structural constraints sort of is that in the general case of a base graph your well inactive inference we're talking about like a Markov chain or some blanket that is kind of ultimately the state space kind of which is included in it if you're trying to map that to a brain or any kind of system with discrete units that are close sometimes and not so close then the arrangement of those will constrain necessarily the base graph to some degree if you if they're right next to each other then there's going to be a greater chance that they are Independence some sort of dependency relationship even if the from like disparate you know far away neurons are actually more strongly correlated they will have to travel through those in some sense the the network there right um so I'm not sure if that's clear but the what I'm saying is that the physical proximity and adjacencies in a physical system kind of partially correlate to something like a markup blanket like the constraints the possible operation in the system yes it can and also this was a major Topic in the recent 51 live streams like we talked about how different by like the pomdp structure partially observable more composition process structure or just generally the active Bayesian graph representation of a generative model that's not the anatomy that would carry out that model you would need to make a neural network with a totally different connectivity to implement that model and so we discussed how do you get from a stated base graph to a neural network architecture that carries out that computation which helps integrate it more closely with experimental results and also software packages and different kinds of computational heuristics and then how do you get from a neural network to a pomdp and so this Paper 51 establishes this relationship formally as well as empirically so it's extremely important reading slash listening in the space of beyond the um more foundational points that have been made in the last several minutes which is just simply these kinds of graphs are not that's just two different categories of visual representations that both use nodes and edges but they're describing different things with nodes and edges and then to go deeper and to understand exactly how biological and artificial

neural architectures implements the kinds of generative models specified by a Bayesian graph then the paper discussed in 51 is the most recent and advanced discussion on it in the field how are categories actually encoded in so can you can you add that reference there oh sure thank you and they've done work in in silico in vitro neurons and in Vivo animals it seems like funny there's fef free energy functional but it's definitely not that and there's been other discussions of just structural versus effective and functional connectivity in all right so how are categories actually encoded in the brain and Neil added a comment I understand that prediction errors are thought to be encoded as pulse density I.E frequency modulated Spike trains as measured on single neurons this might lead us to think that categories are encoded in single neurons like the notorious grandmother and Jennifer Aniston neurons which are neurons that have been reported in different studies to have an increased and sometimes like specific firing rate increase for a specific stimuli so they'll have they'll be measuring thousands of neurons and some neurons fire when some image is brighter or darker and some neurons don't have any clear relationship with visual input but then they'll find some neuron that fires for Jennifer Aniston but not for other closely related stimuli and so hence the fame of the Jennifer Aniston neuron in this schema are we really saying one as far as active is concerned it doesn't matter how categories are stored within a cortical column components we just need to model that they are with associate precisions two we suppose categories are stored statistically somehow collectively within a local population of neurons Would we not expect prediction errors to be represented in Spike trains on many neurons coming into or going out of a cortical column all right there's many ways to explore this question who wants to give a first thought wait on you Ali yeah go for it Ellie oh well actually it's a very big question in the philosophy of mind and the philosophy for cognitive science I mean the question of how uh brain represents the uh the signs and the external objects but one theory is um basically based on versus the taxonomy of signs uh I mean as we um as many were aware uh purse classifies or the signs into three types of symbols icons and indices uh but when we want when we apply this taxonomy in the representation mechanism of brain basically what we need at the very first as as a very first step is to understand how the brain represents indices and icons as a kind of representation processes that would need minimal amount of Consciousness right so uh for indices and icons uh I mean the difference between indices and I icons are that icons are a kind of uh structural structure preserving maps and indices uh are uh we in indices we have a kind of causal relationship between the signs and their reference so before uh um before uh we representing uh the tokens of an object as an index of an external object we need to have a

kind of map or an icon in order to map that in that token of index to our icon right so one theory for doing that is so-called homuncular theory proposed by Daniel Dennett in which he proposes a hierarchy of subjects with decreasing levels of consciousness and on the bottom most level We have basically non-conscious non-conscious subject to somehow grasps the icon in a kind of just a representational map without uh interpreting it as a specific um as a specific Index right but then again uh it passes that representation into its higher levels of hormone homunculus uh with uh increasing their increasing Consciousness until it reaches the subject quote-unquote subject which is basically our whole mind so yeah it's a very big subject and of course it's impossible to summarize it in just a couple of minutes but I would recommend a very interesting book on that subject if anyone wants to further explore that the the first chapter of materialist phenomenology by Manuel enter explains it with a crystal clear prose what that kind of representation or how that mechanism would work nice great answers so a few more angles on this question first the question of representation in active and in Neuroscience is a large area for something to be represented in the brain it doesn't need to be what it is in the outside world clearly but that leads to like a lot of of course um discussion about how representations actually do occur in the brain so the question um was prediction errors are encoded as pulse density as measured on single neurons well the simple answer is it just simply does not have to be that way prediction errors in a different cognitive system or region could be encoded as the autocorrelation function of the spike train or there are neurons that are non-spiking or it could be a brain region there's just no reason to uniquely associate the conveyance of prediction errors to the extent that statistical parameters are even conveyed in neuroanatomy at all which one doesn't even have to agree with but to the extent we're even going to talk about neural phenomena as being related to to statistical phenomena it doesn't have to be the spiking density nor does it have to be on single neurons now that being said in the neural network discussion in 51. there was a relationship that they are showing between the fast perceptual like Dynamics and Spike firing rates and the slower Dynamics and the neural plasticity of the synapses so as kind of before like it could be but just because it could be in in some senses maybe doesn't mean that those are an identical mapping then um well if by that is meant encoded by firing sensitivities in networks of neurons then there is already more being taken into account it's kind of like figure and ground it's not like that neuron taken out of that Network would still signal Jennifer Aniston the semantics of a given category are associated with the inputs and outputs even in cases where there is a statistically tight relationship between the firing activity of some neuron and some other category being

identified um then are we really saying it doesn't matter how categories are stored we just need a model that they are so that is an instrumentalist perspective which is just saying whatever the neurological substrate is of how categories are enacted and represented in the brain we're going to use a specific type of that is super tractable supercomposable translates across systems and so on hashtag active and we're not saying that's the architecture of the system so that's an easy instrumentalist position that can always be taken and we suppose that categories are stored statistically somehow collectively within a local population of neurons that's an empirical hypothesis one could test whether a population rate code or a area of activity or other aspects of neural function that aren't even as commonly discussed depending on the specific question come into play be represented on Spike trains on many neurons coming into and going out of if you expect it if that's your prior then you expect it if it's not your prior you don't expect it there might be anatomical or statistical reasons to scaffold the belief that there would be a many input many output system or not but it can't be answered in the general those are just the examples of Empirical neuroanatomical research to actually pursue um Brock how do you see it related to viviki um I mean relevance realization is uh uh I mean very roughly like opponent processing it's I mean it's I don't I don't really understand the fundamentals of precisely how that would work in a neurological physiological sense but it's kind of again like I guess you could uh like yeah the promotions there um but um yeah that just that you're contrasting like different um fittedness I guess of um in in the live stream he gave this I guess example of um like a dog versus a four-legged animal versus a mammal or whatever why did they say the dog ran by sort of thing um is that that has a good fit and that realized relevance is somehow related to opponent processing um so it's it's that's I guess a way in which it's related to categories or very crudely like General differentiation or something like this tangentially related I don't know yeah I don't know it's like a conceptual approach too yeah there's always different coarse grainings of categories that can be appealed to in speech or in the brain and so outside of contrived you know sorting tasks in the more open-endedness of the niche that the systems have to engage with the categorization [Music] is has different characteristics than like a binary sorting task so that's that's what viviki ties to relevance does anyone want to add another question or thought on five what other neural systems or behaviors are people interested in modeling foreign but also in that in that uh diagram behind the window there there's goals Golds yeah where did those come from seems like um I guess you could do planning without goals but it seems like goals a goal there would be a kind of a reason to plan to get a biological goal of like homeostasis right Isasis is sort of this sort of there was a live

stream you guys did recently on that right some more subjective cultural thing yeah I don't know yeah well here um with the more narrow goal interpretation the difference is between the direct and the indirect policy selection Pathways in the stratum which is mediated by gamma the dopamine signaling and so here on this left side the habits dominate policy selection so e represents the affordances but it's not just like affordances are not just like yes no in the E vector they represent the prior probabilities of given actions so where an affordance is impossible it is given a prior probability of zero and therefore it's never selected where the prior probability on some action affordance being engaged in is high it's more likely to be undertaken so e is not just like a list of what you can choose from it's not like a menu It's like a menu of actions that's weighted by habit in this Construction and so in the absence of any kind of pragmatic or epistemic value which is to say free energy minimization Behavior recapitulates The Habit so there's a mode of policy selection π_i where the entity is merely recapitulating probabilistically their history and habits that's shown on the left side of the stratum then the right side of the stratum shows the cortical engagement seemingly more goal directed whether pragmatic goal directed and or epistemic goal directed or as we think about it in active inference then the pragmatic and epistemic integration into a single imperative G like um let's just say I usually walk up and down but I rarely go left and right and I'm in a maze this left mode of policy selection is just going to be recapitulating the decisions previously made the right mode is going to ideally strike a balance between pragmatic and epistemic value but any discussion of pragmatic and epistemic value so pragmatic value bringing alignments reducing the Divergence between your preferences and observations getting what you want getting what you expect an epistemic value reducing your uncertainty gaining information any discussion of the relative merits of different policies in terms of their expected free energy which again is to say they're relative or relative pragmatic and epistemic value any evaluation of policy is a different cognitive and computational process than merely the recapitulation of habits which doesn't require any consideration of policies at all so this is kind of like walking forward walking into the future looking backwards I mean in some deeper sense aren't we always but this e -mediated mode of decision making does not require any evaluation of counterfactuals or comparison of alternative outcomes now where do habits and preferences come because if under G the pragmatic value is going to be associated with prior probabilities on preferences and um epistemic information as well is going to be mediated by priors which here are being shown as kind of cortically embodied where do those parameters come from Evolution ecology development and behavior over different time scales

so the coarse grained priors are inherited through the process of evolution and then finer and finer scale adjustments can occur via ecological and developmental processes culminating in learning and at the very fast scale perception that's the deflationary approach which would be Evolution has put all those tested processes Evolution ecology development Behavior learning and then perception put that entity's generative model in the context of a generative process into an Adaptive position if it's not adaptive that far from equilibrium system is going to quickly dissipate you just saw something in its last day or it's not its last day and then the inference is going to continue Giuseppe yes something that is not too clear to me habits are supposedly supposedly they do refer uh to things that are learned during the lifetime of the agent as opposed to being deposited through Evolution for instance because one could say also that the the the policies with high pragmatic values that that are the one that preserve the phenotype of the agent are kind of habits but they pertain to the longer time scale of uh phylogenesis so to say so is there the distinction here like the specific of habits the characteristic of web is is that they have been learned they they are they come from a statistical regularities that have been uh accrued and represented through the lifetime of the agent to the experience of the of the agent I believe that it doesn't have to be only learned during the lifetime like let's just say that there's a species of bird that flies at night it habitually flies at night and then you can zoom in more and say now within that bird species they're the early risers and the late risers and this one is habitually Rising early and that one's habitually Rising late now that may be a genetic polymorphism that doesn't even have to be learned at all so I think of course in humans we take a very developmental learning based approach towards the habits and you know Aristotle you know Excellence is what we repeatedly do and all of that but when we just think about habit as what is habitually done then it's a little bit clearer that it's not necessarily learned but then once you couldn't you subsume the like the the tendency to fly more at night as a as a preference that is within the within the yeah as a preference like within the same Matrix yes and in some um convergent evolution scenario we would see the Habit distribution recapitulating the preference distribution increasingly mindlessly offloading the cognitive work of evaluating policies and reducing Divergence through free energy minimization we might expect some of that to become habitualized and merely reflected through the affordances that can be taken or the probability of different affordances in terms of their prior preferences or prior probabilities on affordances um rather than needing to be mediated through preference that's one answer and another kind of corollary answer is on one hand we can talk about preferences over any distribution but in the

bird example the preference for flying at night might not be a category that's accessible to preferences that have to do with um the amount of photons being detected might interact synergistically with habits around like a circadian rhythm or habits around flapping your wings like once the wing is flapped up it always flaps down that doesn't even need to be computationally modeled as a policy decision because it could be relying on these regions of the brain or these kinds of pathways that don't explicitly consider alternatives in the same way and I think again as we discussed at the beginning this isn't the literal interpretation of the neuroanatomy but it's kind of showing two extreme um ends of a continuum um the least to the most explicit consideration of future policies with all the pros and cons of turning the volume up turning the volume down and the volume is computationally analogous to the role of dopamine signaling in the striatum and in insects Ollie uh yeah and uh another slightly different take on this subject is laid out by J Scott Kelso in his book Dynamic patterns the self-organization of brain and behavior uh in which uh he says that because uh the state of uh all the possibilities are too complex to be calculated at each um at each time and for each decision he uh he lays out something like this that if we look at it and look at the state of the possibilities um in terms of its uh distribution of singularities and the basins and The Well of those singularities or or the attractors of uh the Basin of attractors the policy making can be achieved by uh by the effect of the param by the effect of adjusting the parameter and the proximity of those Wells or basins so for example manipulating that parameter parameter to make some wells or basins shallow with the result that the associated preferred state becomes less stable so in this condition random fluctuations actually can spontaneously somehow dislodge the system from a data space that estate and push it into another state so in this case The Habit can also be explained by the varying degrees of these kind of parameter adjustments throughout all the state of possibility States awesome yeah yeah this is this is very interesting one one subject in which I'm I'm interested in applying uh the active inference framework is uh on a meditative practice and actually it has been one proposal is that it may work exactly like that so that uh the the tractors that are very deep and um and are firmly Consolidated through habits through meditative practice becomes shallower also that the range of possibilities of behavior and perception becomes larger because you you you you are not constrained by this deep Wells of uh of uh of energy that have been established through your habits but the interesting thing here the Intriguing part is that through a very repetitive practice kind of repetitive practice you are you some how make your uh energy landscape more shallow at least at least at least during the practice so it's a it's an interesting this paradoxical

aspect how you could somehow train with the very habitual practice to weaken your habits oh just extremely not of a tangent here it's totally interesting and one could imagine real practices or simulated computational agents that engage in repetitive actions and that just sharpens it doesn't open their mind a lot of meditation is not about expanding one's mind or changing one's Behavior it's actually about converging to a certain regime of attention and sharpening distributions that exist during the practice and after the practice there are other cases where we can modulate our receptivity through repetitive actions and being exposed to just merely novelty all the time doesn't necessarily breed novelty in US it might just confuse us mm-hmm yeah Brock some habits are reciprocally narrowing some are reciprocally opening yeah they can look very similar and without um the ability to see the cognitive states of another if somebody's just sitting quietly maybe they're narrowing maybe they're opening maybe they're sleeping well to your point you mean about you know like some basic version of meditation of just like focus on your breathing was like sharpening and narrowing it but in doing that over time you kind of grow this capacity to modulate your attention more on other things more intentionally right right probably not perhaps this idea that by doing that by deploying attention more on the sensory aspect you kind of avoid a little bit or bias the usual drift of precision upwards toward more like Superior layers of the hierarchy associated with mind wandering Etc so this wrong one would you like to add anything awesome I'm going to copy your comment in so we can read this I would put forward here the practice of Alexander technique where we optimize Precision by repetitive action I.E getting in and out of a chair whilst inhibiting and directing hmm yeah it'd be interesting to with this three region neuroanatomical model Oracle the most cognitive most semantically accessible the midbrain striatum most dopamergically associated with this um Continuum between habitual and goal directed policy selection and then action mediated in the peripheral nervous system through um Divergence minimization sketch out I'm sure and then sketch out yeah like um in the Parkinson's case where there's a loss of dopaminergic signaling um which can in the extreme lead to like failures of initiating and planning action um there was a Robin Williams movie Awakenings this movie is about a uh early on Oliver Sacks wrote that book but then there's a Robin Williams movie and um there was a floor of patients in a Medical Institution who were all suffering from basically these different symptoms that today could be recognized as like hypo dopaminergic signaling and then the doctor goes to a conference learns about l-dopa which is like a pre a metabolic precursor of dopamine so dopamine does not cross the blood-brain barrier in mammals however l-dopa does and um then that like awakens people but it

shows like some of the motor and cognitive consequences of hypo and Hyper um one question for me is where does reasoning fit into thicker 5.5 the gears of thought where's the transmission in the carburetor like those are great questions so there's probably many different angles to take on it one angle is like reasoning is a synonym of inference and so by pointing to reasoning we are talking about families of inferential structures and so depending on what that reasoning is about it could manifest in different ways and so like yeah Ali actually helmholz proposes that reasoning can as you said it can be it can be done also unconsciously as inference so uh actually one of the starting points of this whole area Bayesian brain hypothesis and probably its origin is uh helmholtz essay I forgot its exact name from 1866 or something in which he proposes a theory of unconscious inference or unconscious uh conclusion uh in which uh sometimes reasoning actually most of the time the majority of our reasoning processes are not done in our conscious levels so uh yeah I also agree with you that reasoning can be taken as a synonymous for these kind of inferences as well yeah but like on the experiential side it is so fascinating and mysterious stimuli are presented certainly there are um unconscious inferences that facilitate even the initial understanding of a stimuli like a visual stimuli like seeing a letter as a letter seeing a handle for grabbing is kind of the ecological inactivist angle but seeing a symbol or sign and that kind of invokes what Ali described earlier with the theory of semiosis but then we wait and we feel like we reason and we make decisions and so how to how to square the graphical models of inference which is to say reasoning with that yeah Brock I was just gonna ask I guess where mental action fits into this on policy selection or habits or goals like yeah yeah like um I yeah I I think the vast majority of this is yeah that you get like this base unconscious inference but at some point it feels like there's something conscious there about selecting between two things sometimes it really intentional and thinking real hard um so I wonder what that is I guess Giuseppe something about going back to habits something that I just thought about um so usually habitual actions from the point of view from the psychological and phenomenological point of view are they seem to require much less mental energy or psychological energy so we can think about that in terms of uh you know the theory of system one and system two like by then economy and I wonder if they're seeing they seem to have some preferential uh low levels of of free energy also of expected free energy uh the police the habitual policies compared to other policies that somehow mirrors there they're relatively low investment of energy psychologically yeah well that's been a hotly debated topic and one of the most mystifying is that the brain does not change in metabolic activity greatly depending on the task and so

there's all these amazing analogies between Freud and free energy and all these things that Psalms and Fristians and others have worked on and of course the tantalizing connections between free energy minimization and energetic um balance and the um fmri signal that's being studied is the brain oxygen levels changing bold signal and also people can study the glucose metabolism but these levels of change are very small in brain metabolism and glucose usage and they're not simply associated with like quote thinking hard and even necessarily cognitive demands and so it's it's like it's like a I don't know if it's a hammer without a nail or a nail without a hammer because empirically it doesn't change in energy use that much although although there was a recent paper and I think it's in it's on a big Journal I can't remember if it's nature or or actually it was there was you know they proposed some some kind of uh approach similar to energy minimization and and that they were showing that actually the the energy spent in in synaptic communication was was uh minimized and if I I may look for it and I may post it here but I wonder more specifically in this in this idea of that habitual patterns of behavior or let's say sequences of action gets shifted with practice to being process from the cortex to the basal ganglia somehow there is uh there is actually well not not only not only like a metabolic energy like in terms of glucose or Advantage but also in terms of uh really free energy so I don't know if that has been looked at uh specifically like for the basal ganglia or I don't know there must be something because it does seem that mental resources at least experientially are become much the requirement resources become much less once once you a task is automatized let's say yes just shortly to close this and and to address this uh final question when does passive become active inference and end planning well we could look at from a base graph perspective is action involved if the task is merely perceptual we can say that it's passive inference if ocular motor cicading is required so even if the person's in a chair but they're moving their eyeballs that could still be modeled as active inference and you'd want then a unified imperative for inference and action and then when is it planning well as we had some controversial discussions on in cohort one about the second half of the book is it really planning do we really see planning in this textbook um as if planning would be behavior that looks like it's enacting a plan like a chess player who's winning a game but the strict and narrow definition of planning might be explicit consideration of sequential actions and their consequences and preparation for those in evaluation of them and free energy reduction or minimization could be used in purely passive inference which is where it's widely used in general as an Evidence lower bounds and in machine learning without action in the loop so the innovation of active inference is to bring action into the fold of inference and make a unified

free energy functional imperative for perceptual learning and action all right hope it's okay we went a few minutes over but thanks cohort 2 next week we return for some overview and debrief look at the project ideas and also look forward to Future textbook groups so feel free to fill out the um feedback form and encourage others to do so as well and also take a look through some thoughts and add more thoughts on future groups which we'll go over next week so thank you talk to you soon